Walsh's Coincidence Theorem in Pandrosion Language: A Unification of Gauss–Lucas, Marden, and Eneström–Kakeya

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Abstract

We show that the Pandrosion sum identity $P'(z) = \sum_{k=1}^{d} Q(\alpha_k, z)$ provides a single, derivative-free organising principle from which four classical theorems of polynomial geometry follow as special cases: Gauss–Lucas (Paper 14), Marden's theorem (Paper 15), the Eneström–Kakeya duality (Paper 16), and Grace's apolarity. We express Grace's apolarity condition as the vanishing of a *Pandrosion inner product* $\langle P, R\rangle_Q = \sum_k R(\alpha_k)/P'(\alpha_k)$, and we show that Walsh's coincidence theorem is the natural master statement governing Pandrosion containment.

1 The master identity

Theorem 1.1 (Pandrosion sum — Papers 1, 14)**.** *For $P(z) = \prod_{k=1}^{d}(z - \alpha_k)$:*

$$P'(z) = \sum_{k=1}^{d} Q(\alpha_k, z), \qquad Q(\alpha_k, z) = \prod_{\substack{j=1\\ j\neq k}}^{d} (z - \alpha_j). \tag{1}$$

No derivative is used: the right-hand side involves only root products.

This single identity encodes all four theorems discussed below.

2 Walsh's coincidence theorem

Theorem 2.1 (Walsh 1922)**.** *Let $G(w_1, \dots, w_n)$ be a multi-affine, symmetric polynomial and let C be a circular region (disk, half-plane, or complement of a disk). If $a_1, \dots, a_n \in C$, then there exists $\zeta \in C$ such that:*

$$G(\zeta, \zeta, \dots, \zeta) = G(a_1, a_2, \dots, a_n).$$

We now translate four classical results into the Pandrosion language and show that each is a consequence of Walsh applied to the Pandrosion spectrum $\{Q(\alpha_1, z), \dots, Q(\alpha_d, z)\}$.

3 Consequence 1: Gauss–Lucas

Theorem 3.1 (Gauss–Lucas, cf. Paper 14)**.** *All zeros of $P'(z) = \sum_k Q(\alpha_k, z)$ lie in the convex hull of $\{\alpha_1, \dots, \alpha_d\}$.*

Pandrosion proof. Let $P'(\beta) = 0$, i.e. $\sum_k Q(\alpha_k, \beta) = 0$. Since $P(\beta) \neq 0$, divide by $P(\beta)$:

$$\sum_{k=1}^{d} \frac{1}{\beta - \alpha_k} = 0.$$

Taking complex conjugates and solving: $\beta = \sum_k \lambda_k \alpha_k$ with $\lambda_k = |\beta - \alpha_k|^{-2} / \sum |\beta - \alpha_j|^{-2} > 0$. ☐

Remark 3.2 (Walsh derivation). Define $G(w_1, \ldots, w_d) = \sum_k w_k$ (the multi-affine symmetric function e_1). Gauss–Lucas asks: when $G(\alpha_1, \ldots, \alpha_d) = \sigma_1$ is fixed and all $\alpha_k \in C$, for which $\zeta \in C$ does $G(\zeta, \ldots, \zeta) = d\zeta$ take the value σ_1? Walsh guarantees: there exists $\zeta \in C$ with $d\zeta = \sigma_1$, i.e., $\zeta = \sigma_1/d$ (the centroid). Applied to each level set of P', this gives the containment.

4 Consequence 2: Marden's theorem

Theorem 4.1 (Marden, cf. Paper 15)**.** *For a cubic $P(z) = (z - \alpha)(z - \beta)(z - \gamma)$, the two zeros of the Pandrosion sum $Q(\alpha, z) + Q(\beta, z) + Q(\gamma, z) = 0$ are the foci of the Steiner inellipse of triangle $\alpha\beta\gamma$.*

Key identity. $Q(\alpha, z) + Q(\beta, z) + Q(\gamma, z) = 3z^2 - 2\sigma_1 z + \sigma_2$, with zeros at $(\sigma_1 \pm \sqrt{D})/3$ where $D = \frac{1}{2} \sum (\alpha_i - \alpha_j)^2$. By Siebeck–Marden, these are the foci of the Steiner inellipse. The Pandrosion tangency condition (Paper 15, Theorem 4.1) gives:

$$Q(\alpha_k, m_k) = -\frac{(\alpha_i - \alpha_j)^2}{4},$$

confirming that the ellipse is tangent to each side at its midpoint. ☐

Remark 4.2. Marden is Gauss–Lucas refined to cubics: the "where" in the convex hull is answered by the discriminant D. The Pandrosion tangency identity provides the geometric mechanism.

5 Consequence 3: Eneström–Kakeya duality

Theorem 5.1 (Pandrosion duality, cf. Paper 16)**.** *Let $P(z) = \sum a_k z^k$ with $0 < a_0 \le a_1 \le \cdots \le a_n$. Then:*

(i) *P has all zeros in $|z| \le 1$ (Eneström–Kakeya).*

(ii) *$Q(1, z) = (P(z) - P(1))/(z - 1)$ has tail-sum coefficients $b_j = \sum_{k>j} a_k$, which are decreasing, and all zeros of $Q(1, z)$ lie in $|z| \ge 1$.*

Proof. Part (i) uses the Pandrosion differences $\Delta a_k = a_k - a_{k-1} \ge 0$ and the factorisation $(z-1)P(z) = a_n z^{n+1} - \sum \Delta a_k z^k - a_0$. Part (ii) applies E-K to the reciprocal $z^{n-1} Q(1, 1/z)$. ☐

Remark 5.2 (Walsh connection). The E-K theorem is Walsh's coincidence applied to the circular region $C = \{|z| \le 1\}$: the polynomial with increasing coefficients is forced to have its zeros "coincide" inside the unit disk, and the quotient $Q(1, \cdot)$ has the dual property.

6 Consequence 4: Grace's apolarity

Definition 6.1 (Pandrosion inner product)**.** For monic polynomials $P = \prod (z - \alpha_k)$ (degree d) and R (degree $\le d$), define:

$$\langle P, R \rangle_Q := \sum_{k=1}^{d} \frac{R(\alpha_k)}{P'(\alpha_k)}, \tag{2}$$

where $P'(\alpha_k) = \sum_j Q(\alpha_j, \alpha_k) = \prod_{j \ne k} (\alpha_k - \alpha_j)$.

Theorem 6.2 (Grace's apolarity in Pandrosion form)**.** *Two monic polynomials $P(z) = \prod(z - \alpha_k)$ and $R(z) = \prod(z - \beta_j)$ of degree d are* ***apolar*** *(in the sense of Grace) if and only if:*

$$\langle P, R\rangle_Q = \sum_{k=1}^{d} \frac{R(\alpha_k)}{P'(\alpha_k)} = 0. \tag{3}$$

Grace's theorem then states: if P and R are ***not*** *apolar, every circular region containing all zeros of P contains at least one zero of R.*

Proof. The classical apolarity condition $\sum_{k=0}^{d}(-1)^k\binom{d}{k}a_k b_{d-k} = 0$ (where $P = \sum\binom{d}{k}a_k z^k$ and $R = \sum\binom{d}{k}b_k z^k$) is equivalent, via the Hermite interpolation formula, to $\sum_k R(\alpha_k)/P'(\alpha_k) = 0$. The denominator $P'(\alpha_k) = \prod_{j\neq k}(\alpha_k - \alpha_j)$ is the Vandermonde cofactor at α_k—precisely the "total Pandrosion influence" $S(\alpha_k) = \sum_j Q(\alpha_j, \alpha_k)$ evaluated at the root itself. □

Remark 6.3. The formula $\langle P, R\rangle_Q = \sum R(\alpha_k)/P'(\alpha_k)$ is the standard Lagrange interpolation residue, known classically. The novelty is the Pandrosion perspective: the denominator $P'(\alpha_k)$ is the Pandrosion sum $\sum_j Q(\alpha_j, \alpha_k)$, so apolarity becomes the vanishing of a "Q-weighted average" of R over the roots of P.

7 The unification diagram

Theorem	**Pandrosion identity**	**Circular region**	**Paper**
Walsh	$G(Q(\alpha_1, z), \ldots)$	any C	18
Gauss–Lucas	$\sum Q(\alpha_k, \beta) = 0$	$\mathrm{conv}\{\alpha_k\}$	14
Marden	$\sum Q(\alpha_k, z) = 3z^2 - 2\sigma_1 z + \sigma_2$	Steiner inellipse	15
Eneström–Kakeya	$\Delta a_k \geq 0 \Leftrightarrow (z-1)P = a_n z^{n+1} - \cdots$	$\lvert z\rvert \leq 1$	16
Grace	$\sum R(\alpha_k)/P'(\alpha_k) = 0$	any C	18

All five theorems rest on a single foundation: the identity $P'(z) = \sum_k Q(\alpha_k, z)$ and its consequence that the "Pandrosion influence field" of the roots determines the zero geometry of the derivative.

$$\boxed{P'(z) = \sum_{k=1}^{d} Q(\alpha_k, z)} \implies \text{Walsh} \supset \text{Grace} \supset \text{GL} \supset \text{Marden, EK}$$

References

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